

84 66 NFT Deployment Plan (Solana) - Summary

1. Define a structured NFT publication model separating core assets (book), collections (images/music), and granular items (individual songs/images). Organize everything under a single collection with clear hierarchy for scalability and usability.
2. Establish ownership and royalties before minting: author as main creator; resale royalties of 4% for the creator and 1% for the student. Clarify that NFTs do not unlock content directly but represent ownership and access rights.
3. Ensure all source assets are finalized and version-controlled, including book, images, and music files. This is critical because post-mint modifications are complex and should be avoided.
4. Design a standardized metadata schema using JSON following Metaplex conventions. Include fields like name, description, attributes, creators, and links to media stored on Arweave.
5. Use Arweave for decentralized storage and link files through URIs in metadata. This ensures permanence and integrity without storing heavy files directly on-chain.
6. Use compressed NFTs (cNFTs) for large datasets like images and songs to drastically reduce costs. Keep key assets like the book and bundles as standard NFTs for better compatibility and prestige.
7. Structure the system into tiers: core NFTs (book), bundle NFTs (chapters), and atomic NFTs (images/songs). This layered approach improves organization and flexibility.
8. Estimate required SOL: devnet is free, while mainnet costs are low (~0.2–1 SOL total). This includes minting and transaction fees depending on NFT types. Costs are entirely funded by the author/main creator in all cases.
9. Create a main collection NFT using Metaplex and assign all assets to it. This enables grouping, verification, and easier discovery in wallets and marketplaces.
10. Automate metadata JSON generation using scripts (Java or Node.js). This avoids manual errors and supports scaling for hundreds of NFTs.
11. List NFTs on marketplaces like Magic Eden or Tensor to improve visibility and liquidity. This is essential for adoption and trust among users.
12. Develop minting scripts or backend using Metaplex SDK to upload assets, create metadata, mint NFTs, and assign them to collections programmatically.

13. Perform collection verification and manage authorities to ensure authenticity and control. This prevents spoofing and maintains project integrity.
14. Create a full-cycle demo including wallet connection, NFT acquisition, and token-gated content access. This demonstrates real-world usability for the thesis.
15. Implement token-gated access by storing previews publicly and serving full content only to NFT holders via a web platform. This enables controlled distribution.
16. Adopt a hybrid edition strategy: unlimited editions for book/music but limited editions for images to introduce scarcity and increase value.
17. Complement blockchain with legal protections by defining licenses and copyright terms. Smart contracts do not replace legal ownership.
18. Limit master thesis scope to a manageable subset (book + 20 images + 10 songs). This is sufficient to validate the architecture without overcomplicating.
19. Test thoroughly on Solana devnet before deploying on mainnet. This ensures reliability and prevents costly errors.
20. Estimate total costs: master scope ~5–10€, full project ~15–30€ plus storage costs. Arweave may add additional expenses depending on file size.

The following diagram illustrates the full workflow from asset preparation to NFT minting, storage, and distribution.

Description:

- Assets are prepared and converted into metadata.
- Files and metadata are uploaded to Arweave.
- Minting scripts create NFTs on Solana and assign them to a collection.
- NFTs are distributed via marketplaces and accessed through a token-gated website.

